

Kinora Rib Cage & Thorax

Clinical AI Orientation (Part 13)

Ribs, Sternum, Costal Joints, Breathing, TOS & Injuries

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

Educational only. Chest pain — exclude cardiac/PE/pneumothorax emergently. Flail chest and vascular TOS complications are emergencies.

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1. Bones

RECORD: Ribs 1–12 Overview

ID: RIB-BONE-01

Structure: 12 pairs; 1–7 true, 8–10 false, 11–12 floating

Landmarks: Head, neck, tubercle, angle, costal groove (VAN)

Articulations: Costovertebral, costotransverse, costochondral/sternocostal

Clinical: Rib fracture, flail chest, cervical rib TOS, slipping rib (8–10)

Imaging: CXR, CT trauma

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sternum

ID: RIB-BONE-02

Structure: Manubrium, body, xiphoid

Landmarks: Jugular notch, sternal angle (T4–5 / 2nd costal cartilage)

Clinical: Sternal fracture, manubriosternal junction, bone marrow biopsy site

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Costal Cartilage

ID: RIB-BONE-03

Structure: Hyaline cartilage ribs to sternum / interchondral

Clinical: Costochondritis, Tietze syndrome

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

2. Joints

RECORD: Costovertebral Joint

ID: RIB-JNT-01

Type: Synovial plane (head of rib—vertebral body demifacets)

Motion: Pump/bucket handle contributions

Clinical: Hypomobility with thoracic pain; trauma

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Costotransverse Joint

ID: RIB-JNT-02

Type: Synovial (tubercle—TP)

Motion: Rotation gliding with respiration

Clinical: Manual therapy target; 1st rib elevation TOS

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

3. Muscles

RECORD: Diaphragm

ID: RIB-MUS-01

Origin: Xiphoid, costal margin, lumbar crura

Insertion: Central tendon

Innervation: Phrenic C3–5

Function: Primary inspiratory muscle; IAP/canister with pelvic floor/TA

Clinical: Paradoxical breathing, phrenic palsy, referral shoulder (Kehr)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: External/Internal Intercostals

ID: RIB-MUS-02

Function: External: inspiration assist; Internal: expiration assist

Innervation: Intercostal nerves T1–T11

Clinical: Strain, herpes zoster along dermatome

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Pectoralis Major/Minor

ID: RIB-MUS-03

Function: Arm motion; pec minor depresses scapula / can compress brachial plexus (TOS)

Clinical: Pec minor TOS, strain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

4. Biomechanics

RECORD: Breathing Mechanics

ID: RIB-BIO-01

Pump_Handle: Upper ribs — AP diameter increase

Bucket_Handle: Lower ribs — transverse diameter

Diaphragm: Dome descent increases vertical dimension; zone of apposition

Clinical: Dysfunctional breathing with neck pain/anxiety

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Thoracic Outlet

ID: RIB-BIO-02

Spaces: Interscalene triangle, costoclavicular space, subcoracoid/pec minor

Contents: Brachial plexus, subclavian artery/vein

Clinical: NTOS/ATOS/VTOS differentials; effort thrombosis urgent

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

5. Pathologies

RECORD: Rib Fracture / Flail Chest

ID: RIB-PATH-01

Red_Flag: Respiratory compromise, pneumothorax, flail — emergency

Rehab: Pain control, breathing, incentive spirometry, early mobility

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Costochondritis / Tietze

ID: RIB-PATH-02

Symptoms: Reproducible costal cartilage tenderness; Tietze has swelling

DDx: ACS cardiac — always screen cardiac red flags for chest pain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: 1st Rib / TOS Related Pain

ID: RIB-PATH-03

Assessment: Elevated 1st rib, Roos/Adson interpret cautiously (limited Sn/Sp)

Treatment: Scapular PT, breathing, neural mobility; vascular urgent if ischemic

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.